Lecture 2: Foundations: This Is Your Brain

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-2 (CC BY-NC-SA 3.0).

Before you start

This lecture asks how the mind relates to the body. Professor Bloom compares dualism, the common-sense idea that the mind is separate from the body, with the scientific view that "the mind is what the brain does." Then he gives a basic tour of the brain that you will need for the rest of the course.

Learning goals

After this lecture you should be able to:

- Explain the difference between dualism and the view that the mind is what the brain does.
- Describe Descartes' two arguments for dualism and the reasons why scientists now reject dualism.
- Describe how a neuron works and how neurons communicate through neurotransmitters.
- Name the main parts of the brain and give an example of what we learn from brain damage.
- Explain what the "hard problem" of consciousness is.

Key ideas

1. The Brain, the Mind and Dualism

- Francis Crick's "Astonishing Hypothesis" says that everything you are, including your memories and free will, is "no more than the behavior of a vast assembly of nerve cells and their associated molecules."
- Most people are instead **dualists**: they believe the mind is separate from the body. The best-known defender of dualism was the philosopher Rene Descartes. He said animals are "beast machines," but humans are immaterial souls that have physical bodies.
- Descartes' first argument: machines (like the water-powered robots in the French Royal Gardens) only react in fixed ways, but humans can choose, for example in how they use language.
- His second argument used the **method of doubt**. He could doubt almost everything, even that he had a body; maybe an evil demon was tricking him (the modern version is *The Matrix*). But he could not doubt that he was thinking. So, he said, the mind is different from the body.
- Dualism is common sense. It shows in our language ("my body," "my brain"), in stories where a person wakes up in another body (Kafka's Gregor Samsa becomes an insect), and in the common belief that people survive the death of their body.

2. Scientific Consensus against Dualism

- The scientific consensus is that dualism is wrong: "the mind is what the brain does," just as computation is what a computer does.

- Reason 1: dualism is unscientific. Saying "it's all nonphysical" explains nothing, and dualists cannot explain how a physical body connects to an immaterial soul.
- Reason 2: we now have an "existence proof." Machines can play chess and manipulate symbols, so physical things can do complicated things.
- Reason 3: the brain is clearly involved in mental life. Head injuries, diseases and chemicals like alcohol change how we think. Imaging methods (CAT scans, PET and fMRI) show different parts of the brain active during different kinds of thinking.

3. The Neuron: The Basic Building Blocks of Thought

- The basic unit of the brain is the **neuron**. Its **dendrites** receive signals. Excitatory signals make firing more likely; inhibitory signals make it less likely. The cell body adds them up, and if the total is high enough, the neuron fires along its **axon**. A **myelin sheath** (insulation) makes firing quicker.
- There are about one thousand billion neurons (the figure given in the 2007 lecture; today's best estimate is about 86 billion). Sensory neurons take in information, motor neurons tell muscles what to do, and interneurons connect the two and "do the thinking."
- A neuron is "all or nothing," like a gun. Intensity is coded by the number of neurons firing and by how often they fire.
- Neurons communicate chemically. At the **synapse** (a tiny gap), the axon releases **neurotransmitters**. **Agonists** are drugs that increase the effect of neurotransmitters; **antagonists** reduce it. Curare is an antagonist: it blocks motor neurons from affecting muscles and causes paralysis.
- The brain survives damage well and is very fast, even though tissue is slow. So it is not wired like a personal computer; it uses **massively parallel distributed processing**. Researchers build "neural networks" to copy this, but at the time of the lecture (2007) no machine could recognize faces or understand sentences at the level of a two-year-old.

4. The Different Parts of the Brain

- Some things, like reflexes, need no brain. A book dropped on a student's hand made the hand go back up automatically.
- **Subcortical structures** lie below the cortex: the medulla (heart rate and breathing), the cerebellum (balance and coordination) and the hypothalamus (feeding, hunger, thirst and partly sleep).
- The **cortex** is the crumpled outer layer, about 80% of the volume of our brain. It has four lobes: frontal, parietal, occipital and temporal.
- Dr. Penfield at McGill University stimulated people's brains with electricity during surgery. This revealed **maps of the body** in the motor cortex and the somatosensory cortex. The maps are **topographical**: body parts close together on the body are close together in the brain. Parts with more control or sensation, like the tongue and face, get more space.
- Brain damage also teaches us. **Apraxia**: you can move but cannot coordinate movements, like waving goodbye. **Agnosia**: you can see but cannot recognize things; **prosopagnosia** is losing the

ability to recognize faces. **Aphasia**: a language disorder (Paul Broca's patient in 1861 could only say "tan").

5. Mechanist Conception and the Hard Problem of Consciousness

- The two halves (hemispheres) of the brain are not the same. In right-handed people (about nine out of ten people), language is in the left hemisphere; math and music are on the right.
- There is a crossover: the right hemisphere controls the left side of the body, and the other way around. The halves are connected by the **corpus callosum**. A film about Michael Gazzaniga's work showed a man whose two hemispheres did not communicate.
- The **"hard problem" of consciousness**: a computer can play chess, but it does not have feelings. How does a physical brain give rise to subjective experience? Huxley compared it to the genie appearing "when Aladdin rubs his lamp."
- The course takes a **mechanistic** view: it tries to explain the mind. This creates tension with ideas like free will. There are three choices: reject the science, reject these humanist values, or try to reconcile them.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
dualism	The belief that the mind and the body are two separate things	Descartes: humans are souls that have bodies
existence proof	A real example that shows something is possible	Machines can play chess
neuron	A nerve cell; the basic unit of the brain	Billions in the brain (86 billion by today's estimate)
synapse	The tiny gap between two neurons	Neurotransmitters cross it
neurotransmitter	A chemical that carries a signal between neurons	Serotonin, dopamine
agonist / antagonist	A drug that increases / reduces the effect of neurotransmitters	Curare is an antagonist
cortex	The wrinkled outer layer of the brain	About 80% of the volume of our brain
topographical	Arranged like a map	The tongue is near the jaw in the motor cortex
agnosia	Seeing well but not recognizing things	Prosopagnosia: not recognizing faces
aphasia	A language disorder after brain damage	Broca's patient who could only say "tan"
corpus callosum	The connection between the two halves of the brain	The man in the film whose hemispheres did not communicate

Term	Meaning in simple English	Example from the lecture
hard problem of consciousness	The puzzle of how a brain produces feelings and experience	Huxley's genie and Aladdin's lamp

Check your understanding

1. What is dualism, and who was its best-known defender?
2. Describe Descartes' method of doubt. What could he not doubt?
3. Give two reasons why scientists now reject dualism.
4. A neuron is "all or nothing." How can we still feel different strengths of touch or heat?
5. What is the difference between an agonist and an antagonist? Give one example.
6. Why is the tongue so large in the brain's sensory map?
7. How does brain damage help us learn what parts of the brain do? Give an example.
8. What is the "hard problem" of consciousness?

Answer guide

1. Dualism is the belief that the mind is separate from the physical body. Its best-known defender was Rene Descartes.
2. He asked what he could be sure of. He could doubt even his body, because he might be tricked or dreaming. But he could not doubt that he was thinking.
3. Dualism cannot explain how a soul connects to a body. Also, machines show that physical things can do complex things, and brain activity clearly matches mental life.
4. Intensity is coded by the number of neurons firing and by how often they fire.
5. An agonist increases the effect of neurotransmitters; an antagonist reduces it. Curare is an antagonist.
6. Size in the brain map depends on how much sensation or control there is, not on real size. The tongue has many sense organs.
7. If damage to one area causes a specific problem, that area is probably linked to that ability, for example recognizing faces.
8. It is the puzzle of how a physical brain gives rise to subjective experience and feelings.

English practice

Useful phrases from the lecture

- "It turns out that..." - introduces a surprising fact. Example: "It turns out that neurons relate to one another chemically."
- "It used to be believed that..." - describes an old idea that is now rejected. Example: "It used to be believed that the brain is wired up like a computer."
- "This is no surprise if..., but it is extremely difficult to explain if..." - compares two theories.
- "The scientific consensus is that..." - says what most scientists agree on.
- "Faced with this tension, there are three possibilities." - introduces options for a problem.

Speaking task (with your AI tutor)

- Explain how a neuron fires and sends a signal in 4-6 sentences. Use these terms: *dendrites*, *axon*, *synapse*.
- Explain why most scientists reject dualism in 4-6 sentences. Use these terms: *existence proof*, *brain damage*, *fMRI*.

Writing task

- Write one paragraph (80-120 words): Describe Descartes' two arguments for dualism. Then explain one reason why modern scientists reject his conclusion.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 2. Ask me questions about dualism and Descartes, one at a time."
- "Quiz me on the key vocabulary of Lecture 2."
- "I will explain how neurons communicate through neurotransmitters. Check my understanding and my English."
- "Ask me to describe agnosia and aphasia."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 133-167
- Damasio, Antonio R. "Descartes' Error: Emotion, Reason, and the Human Brain." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 58-69
- Sacks, Oliver. "The Man Who Mistook His Wife for a Hat and Other Clinical Tales." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 70-79

My notes